

this is true for the classical Tate curve. To glue each  $\widehat{T}$  with the rational Tate curve  $T_{\mathbf{Q}}$ , we will use an arithmetic fracture square, so the Cartesian diagram of  $\mathbf{E}_{\infty}$ -rings

$$\begin{array}{ccc} \mathbf{B} & \longrightarrow & \prod_p \widehat{\mathbf{B}} \\ \downarrow & & \downarrow \\ \mathbf{B}_{\mathbf{Q}} & \longrightarrow & \left( \prod_p \widehat{\mathbf{B}} \right)_{\mathbf{Q}}. \end{array}$$

As the functor  $\mathrm{Ell}^{\mathrm{or}}(-)^{\simeq} : \mathrm{CAlg} \rightarrow \mathcal{S}$  is corepresented by  $\mathrm{TMF}$  when restricted to complex periodic  $\mathbf{E}_{\infty}$ -rings by Th.5.4, we have equivalences of spaces

$$\begin{aligned} \mathrm{Ell}^{\mathrm{or}}(\mathbf{B})^{\simeq} &\simeq \mathrm{Map}_{\mathrm{CAlg}}(\mathrm{TMF}, \mathbf{B}) \simeq \mathrm{Map}_{\mathrm{CAlg}}(\mathrm{TMF}, \mathbf{B}_{\mathbf{Q}}) \times_{\mathrm{Map}_{\mathrm{CAlg}}(\mathrm{TMF}, (\prod_p \widehat{\mathbf{B}})_{\mathbf{Q}})} \mathrm{Map}_{\mathrm{CAlg}}(\mathrm{TMF}, \prod_p \widehat{\mathbf{B}}) \\ &\simeq \mathrm{Ell}^{\mathrm{or}}(\mathbf{B}_{\mathbf{Q}})^{\simeq} \times_{\mathrm{Ell}^{\mathrm{or}}((\prod_p \widehat{\mathbf{B}})_{\mathbf{Q}})^{\simeq}} \mathrm{Ell}^{\mathrm{or}}(\prod_p \widehat{\mathbf{B}})^{\simeq}. \end{aligned}$$

We already have the desired oriented elliptic curves  $\widehat{T}$  over each  $\widehat{\mathbf{B}}$  and  $T_{\mathbf{Q}}$  over  $\mathbf{B}_{\mathbf{Q}}$ , so to obtain  $T$  over  $\mathbf{B}$ , it suffices to produce an isomorphism between oriented elliptic curves over  $(\prod_p \widehat{\mathbf{B}})_{\mathbf{Q}}$ . To do this, consider the equivalences of spaces

$$\begin{aligned} \mathrm{Ell}^{\mathrm{or}}\left(\left(\prod_p \widehat{\mathbf{B}}\right)_{\mathbf{Q}}\right)^{\simeq} &\simeq \mathrm{Map}_{\mathrm{CAlg}}\left(\mathrm{TMF}, \left(\prod_p \widehat{\mathbf{B}}\right)_{\mathbf{Q}}\right) \simeq \mathrm{Map}_{\mathrm{CAlg}_{\mathbf{Q}}}\left(\mathrm{TMF}_{\mathbf{Q}}, \left(\prod_p \widehat{\mathbf{B}}\right)_{\mathbf{Q}}\right) \\ &\subseteq \mathrm{Map}_{\mathrm{CAlg}_{\mathbf{Q}}}\left(\mathrm{tmf}_{\mathbf{Q}}, \left(\prod_p \widehat{\mathbf{B}}\right)_{\mathbf{Q}}\right) \simeq \Omega^{\infty+8}\left(\left(\prod_p \widehat{\mathbf{B}}\right)_{\mathbf{Q}}\right) \times \Omega^{\infty+12}\left(\left(\prod_p \widehat{\mathbf{B}}\right)_{\mathbf{Q}}\right), \end{aligned} \quad (5.5)$$

where we use that  $\mathrm{tmf}_{\mathbf{Q}}$  is the free  $\mathbf{E}_{\infty}$ - $\mathbf{Q}$ -algebra on two generators  $c_4$  and  $c_6$  in degrees 8 and 12, respectively, that  $\mathrm{TMF}_{\mathbf{Q}} \simeq \mathrm{tmf}_{\mathbf{Q}}[\Delta^{-1}]$  for  $\Delta = (c_4^3 - c_6^2)/1728$ , and that the containment above is a monomorphism onto those path components where the image of  $\Delta$  is invertible. Both the base-change of  $\widehat{T}$  and  $T_{\mathbf{Q}}$  to  $(\prod_p \widehat{\mathbf{B}})_{\mathbf{Q}}$  are chosen such that the associated map from  $\mathrm{tmf}_{\mathbf{Q}}$  sends  $c_4$  and  $c_6$  to their  $q$ -expansion maps. From the equivalences above, this shows that these oriented elliptic curves are equivalent over  $(\prod_p \widehat{\mathbf{B}})_{\mathbf{Q}}$ . The fact that  $(\prod_p \widehat{\mathbf{B}})_{\mathbf{Q}}$  has homotopy groups concentrated in even degrees also shows that the choice of equivalence is unique up to homotopy; even up to 3-homotopy. This yields the oriented elliptic curve  $T$  over  $\mathbf{B}$ .

To compute the formal group associated with  $T$ , notice that  $\mathbf{B}$  is an  $\mathbf{E}_{\infty}$ -KU-algebra, as it then follows that the formal group associated with  $T$  is  $\widehat{\mathbf{G}}_m$ . Indeed, by [Lur18b, Pr.4.3.23], an orientation on  $T$  is an equivalence between its associated formal group and the Quillen formal group over  $\mathbf{B}$ . As the Quillen formal group of a complex-periodic  $\mathbf{E}_{\infty}$ -ring satisfies base-change, we obtain the desired result of the Quillen formal group  $\widehat{\mathbf{G}}_m$  over KU. Alternatively, one can use the fact that the space of oriented formal groups over a complex-periodic